

Table 5 Critical values for the χ^2 -distributionColumns a denote the lower boundaries or the left-sided critical values.Columns b denote the upper boundaries or the right-sided critical values.

	Level of significance α							
Two-sided One-sided	0.20 0.10		0.10 0.05		0.05 0.025		0.01 0.005	
ν	a	b	a	b	a	b	a	b
1	0.016	2.71	39.10^{-4}	3.84	98.10^{-5}	5.02	16.10^{-5}	6.63
2	0.21	4.61	0.10	5.99	0.05	7.38	0.02	9.21
3	0.58	6.25	0.35	7.81	0.22	9.35	0.11	11.34
4	1.06	7.78	0.71	9.49	0.48	11.14	0.30	13.28
5	1.61	9.24	1.15	11.07	0.83	12.83	0.55	15.09
6	2.20	10.64	1.64	12.59	1.24	14.45	0.87	16.81
7	2.83	12.02	2.17	14.07	1.69	16.01	1.24	18.48
8	3.49	13.36	2.73	15.51	2.18	17.53	1.65	20.09
9	4.17	14.68	3.33	16.92	2.70	19.02	2.09	21.67
10	4.87	15.99	3.94	18.31	3.25	20.48	2.56	23.21
11	5.58	17.28	4.57	19.68	3.82	21.92	3.05	24.73
12	6.30	18.55	5.23	21.03	4.40	23.34	3.57	26.22
13	7.04	19.81	5.89	22.36	5.01	24.74	4.11	27.69
14	7.79	21.06	6.57	23.68	5.63	26.12	4.66	29.14
15	8.55	22.31	7.26	25.00	6.26	27.49	5.23	30.58
16	9.31	23.54	7.96	26.30	6.91	28.85	5.81	32.00
17	10.09	24.77	8.67	27.59	7.56	30.19	6.41	33.41
18	10.86	25.99	9.39	28.87	8.23	31.53	7.01	34.81
19	11.65	27.20	10.12	30.14	8.91	32.85	7.63	36.19
20	12.44	28.41	10.85	31.41	9.59	34.17	8.26	37.57
21	13.24	29.62	11.59	32.67	10.28	35.48	8.90	38.93
22	14.04	30.81	12.34	33.92	10.98	36.78	9.54	40.29
23	14.85	32.01	13.09	35.17	11.69	38.08	10.20	41.64
24	15.66	33.20	13.85	36.42	12.40	39.36	10.86	42.98
25	16.47	34.38	14.61	37.65	13.12	40.65	11.52	44.31
26	17.29	35.56	15.38	38.89	13.84	41.92	12.20	45.64
27	18.11	36.74	16.15	40.11	14.57	43.19	12.88	46.96
28	18.94	37.92	16.93	41.34	15.31	44.46	13.56	48.28
29	19.77	39.09	17.71	42.56	16.05	45.72	14.26	49.59
30	20.60	40.26	18.49	43.77	16.79	46.98	14.95	50.89
40	29.05	51.81	26.51	55.76	24.43	59.34	22.16	63.69
50	37.69	63.17	34.76	67.50	32.36	71.42	29.71	76.15
60	46.46	74.40	43.19	79.08	40.48	83.30	37.48	88.38
70	55.33	85.53	51.74	90.53	48.76	95.02	45.44	100.43
80	64.28	96.58	60.39	101.88	57.15	106.63	53.54	112.33
90	73.29	107.57	69.13	113.15	65.65	118.14	61.75	124.12
100	82.36	118.50	77.93	124.34	74.22	129.56	70.06	135.81